

Vrund Patel

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Education

BE: Vishwakarma Government Engineering College 2022 – 2026 | **CGPA: 9.21**
Computer Science & Engineering (Data Science)

HSC: Swastik Education Campus (GSEB Board) 2022 | **Percentage: 88.67%**

Experience

AI/ML Engineer Intern – iQudTek 📅 May 2026 – Present

- Built and shipped an end-to-end computer vision pipeline for pharmaceutical smoke-study analysis – scene gating, YOLO-seg smoke detection (**0.879 mAP@50**), pipe/camera validation gates, optical flow, and statistical laminar baseline scoring – now powering the company's core product and client deliverables.
- Cut smoke-study review from **2–3 hours** across 3 QA engineers to a **13-minute** automated run, with a 10-metric deviation engine delivering explainable verdicts and annotated defect clips, and a feedback loop reducing repeat false positives by **92%**.
- Deployed GPU inference on SageMaker with an S3 job queue and Lambda triggers for a fully automated upload-to-results workflow, instrumented with CloudWatch for monitoring and cost control.
- Independently owned the full lifecycle: data annotation, model training, pipeline architecture, cloud deployment, and production integration.

AI/ML Engineer Intern – Bacancy Technology 📅 Jan 2026 – May 2026

- Completed 15+ hands-on implementations across ML, DL, CV, NLP, and GenAI — including CNNs, RNNs/LSTMs, and transformer-based models — covering data preprocessing, model training, and evaluation.
- Designed workflow automations with n8n to streamline business processes and API integrations.
- Built scalable backend services using FastAPI and integrated them with React applications.

Projects

RoastForge – MCP-Enabled Agentic AI Resume Roaster & Rebuilder [🔗 RoastForge](#) [🌐 Video](#)

- Built an 8-node LangGraph + FastMCP agentic workflow that parses resumes, brutally roasts weaknesses, evaluates ATS fit (0–100), iteratively rebuilds until score > 90, generates interview questions, and exports professional PDF resumes autonomously.
- Integrated Claude Desktop MCP with parallel node execution (Roast + ATS), conditional routing loops, MCP Resources for real PDF delivery, and structured Pydantic outputs; Increased average ATS score from 55 to 84 (+53%) through iterative multi-agent refinement.

Agentic RAG for Code-Mixed WhatsApp Chat Memory [🔗 WhatsUp](#) [🌐 Video](#)

- **Hybrid retrieval** - dense (bge-m3) + BM25, RRF fusion, bge-reranker-v2-m3 cross-encoder reranking; per-chunk LLM glosses to make romanized Gujarati embeddable.
- **LangGraph pipeline** - LLM emits a structured filter (intent / date range / sender), deterministic routing to semantic | temporal | event nodes.
- **Voice + streaming** - Whisper large-v3-turbo (Groq) speech-to-text with a transcript-confirmation gate, offline faster-whisper fallback, SSE token streaming with live pipeline trace, Ollama & Groq swappable per task.

Technical Skills

Languages: Python, SQL

GenAI & Agentic AI: LangChain, LangGraph, LangSmith, RAG, MCP, FAISS, Hybrid Retrieval, Reranking, Fine-Tuning, Prompt Engineering, Hugging Face

Machine Learning & Deep Learning: Scikit-learn, PyTorch, TensorFlow, Keras, CNNs, LSTMs, Transformers, NLP, Feature Engineering, Model Evaluation

Computer Vision: OpenCV, YOLO, Roboflow, Object Detection, Image Segmentation, Optical Flow

Cloud & MLOps: AWS (SageMaker, Lambda, S3, EC2, CloudWatch, IAM), Git, GitHub

Tools & Frameworks: FastAPI, React, n8n, Pandas, NumPy, Matplotlib, Power BI, Jupyter